Chapter 1: Survival analysis — Solutions

STA310 Survival Analysis (Master's in Applied Biostatistics)

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Chapter 1 sets up the objects of survival analysis — censoring, the time origin and time scale, and the survivor, hazard and cumulative hazard functions. These exercises check that you can reason about censoring and the time scale, and interpret $S(t)$, $h(t)$ and $H(t)$ in concrete medical settings.

1 Special features of survival data (Book §1.2)

1.1 Censoring in a clinical trial

Type: Concept & notation

A trial follows patients from surgery until recurrence of a tumour, up to a maximum of 24 months. The table gives the follow-up time (months) and what was observed for each patient.

Patient	Follow-up (months)	Outcome
1	8	recurrence
2	20	recurrence
3	24	no recurrence by end of study
4	15	lost to follow-up
5	24	no recurrence by end of study
6	12	recurrence

(a) Which patients' times are censored, and which are observed (uncensored)?
(b) For patient 4 (lost to follow-up at 15 months), what can we say about their time to recurrence?
(c) Why can we not simply discard the censored patients when summarising the data?

Solution.

(a) Censored: patients 3, 4 and 5 (no event was observed). Observed: patients 1, 2 and 6 (recurrence at a known time).
(b) We only know that the recurrence, if any, occurred after 15 months; the exact time is unknown.
(c) Discarding them throws away the information that they were event-free up to their censoring time. Only the patients who recurred (mostly early) would remain, so the average time to recurrence would be biased downward.

1.2 Study time and patient time

Type: Concept & notation

A clinical trial recruits patients over a two-year period and follows each of them until a common closing date, after which no further information is collected.

(a) Explain the difference between *study time* (calendar time) and *patient time* (time since the patient's time origin) in this trial.
(b) For a patient recruited one year after the trial starts, how does their patient time relate to study time?
(c) Why is the analysis of survival data usually based on patient time rather than study time?

Solution.

(a) Study time is measured on the calendar, over the period the individual is in the study; patient time is measured from that individual's own time origin (entry) to the event or to censoring.
(b) At any calendar date, that patient's study time equals their patient time plus one year.
(c) Patient time gives every individual a common, meaningful origin (their entry into the study), so the resulting survival times are comparable; study time would confound different recruitment dates with the follow-up already elapsed.

1.3 Choosing the time origin

Type: From research question to model

A study investigates the survival of patients with a serious disease. Patients are diagnosed at different calendar dates, and treatment may start some weeks after diagnosis. During follow-up, some patients die while others are still alive when the study ends.

(a) Propose a sensible time origin for measuring survival time, and explain your choice.
(b) Name one alternative time origin that could be used, and describe how the choice of time origin changes the question the analysis answers.
(c) What is the endpoint (the event) in this study?

Solution.

(a) A common choice is the date of diagnosis (or the date treatment starts). The time origin should be a well-defined, comparable starting point for all patients.
(b) For example, the date treatment starts. Survival measured from diagnosis answers a different question than survival measured from the start of treatment, and the two origins can also change which patients are comparable.
(c) The endpoint is death; patients still alive at the end of follow-up are censored.

1.4 Independent censoring

Type: Assumptions

(a) Explain in your own words what *independent* (non-informative) censoring means.
(b) Give an example of a study where censoring would NOT be independent, and explain why it would then be misleading to ignore the censoring.
(c) Why is independent censoring needed for a valid analysis of censored survival data?

Solution.

(a) The censoring mechanism is unrelated to the subject's risk of the event: an individual censored at time t is representative of all individuals who have survived to t.
(b) For example, patients withdrawn from a trial because their condition has deteriorated are censored *because* their prognosis is poor. Those who remain at risk are then healthier than the population, so the results would be biased.
(c) If censoring is informative, those who remain at risk are not representative of the population, so the apparent survival experience of the group is distorted and any conclusions would be misleading.

1.5 A misleading analysis

Type: What's wrong here?

A researcher studying time to recurrence excludes from the analysis all patients who had not yet had a recurrence when the study ended, and then reports the average time to recurrence among the remaining patients.

Explain what is wrong with this approach and how it distorts the result.

Solution. The excluded patients are censored: they were event-free up to the end of the study and carry information. By dropping them, only the patients who did recur (mostly earlier) remain, so the average time to recurrence is underestimated (biased downward) and the survival experience of the group is misrepresented. Censored observations must be handled, not discarded.

2 Survivor, hazard and cumulative hazard functions (Book §1.4)

2.1 Relating the survival and hazard functions

Type: Concept & notation

(a) How does the cumulative hazard $H(t)$ relate to the survivor function $S(t)$?

(b) The hazard is $h(t) = -\frac{\mathrm{d}}{\mathrm{d}t} \log S(t)$. Using this, express $H(t)$ in terms of $h(t)$, and explain in words what $H(t)$ accumulates.

Solution.

(a) $S(t) = \exp\{-H(t)\}$, equivalently $H(t) = -\log S(t)$. As $S(t)$ falls from 1 toward 0, $H(t)$ grows from 0 upward.
(b) $H(t) = \int_0^t h(u)\,\mathrm{d}u$: the cumulative hazard adds up the hazard (the total risk experienced) from time 0 up to time t.

2.2 Interpreting the hazard

Type: Concept & notation

Suppose that, for patients one year after a surgical procedure, the hazard of death is 0.2 per year.

(a) Explain in words what a hazard of 0.2 per year means.
(b) Explain why it is not correct to say that "20% of patients die during the next year".
(c) Can the hazard take values greater than 1? What about the survivor function $S(t)$?

Solution.

(a) The hazard is the instantaneous rate of death per year among patients who have survived to one year — 0.2 events per patient-year at risk, conditional on being alive at that time.
(b) The hazard is a rate, not a probability. The proportion dying within the next year is $1 - S(1)$; for a constant hazard of 0.2 this is $1 - e^{-0.2} \approx 0.18$, not 0.2. A rate is conditional on survival and is not bounded by 1, whereas a probability is.
(c) Yes — the hazard is a rate and can exceed 1. $S(t)$ is a probability and always lies between 0 and 1.

2.3 True or false

Type: True / false with justification

State whether each claim is true or false and justify your answer in one sentence.

(a) The hazard function can take values greater than 1.
(b) If a subject is censored at time t, we know the event occurred after t.
(c) $S(t)$ is a non-increasing function of t.

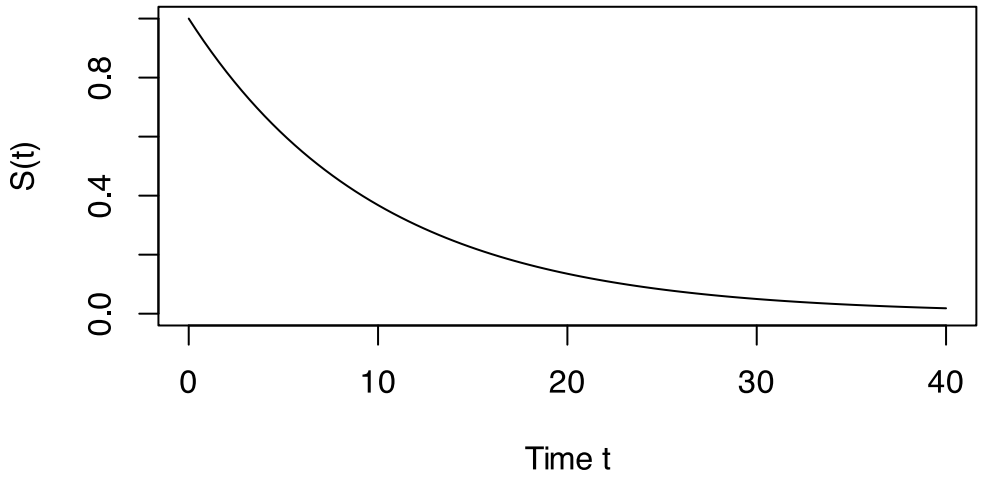
Solution.

(a) True — the hazard is a rate, not a probability, so it is not bounded by 1.
(b) True — right-censoring at t means the subject was still event-free at t; the event, if any, occurs after t (under independent censoring).
(c) True — as time increases, $S(t)$ can only stay constant or decrease.

2.4 Interpreting the survivor function

Type: Interpret figure

The figure shows the survivor function $S(t) = e^{-0.1t}$ for a population with constant hazard $\lambda = 0.1$.



(a) Read off the values of $S(10)$ and $S(20)$ approximately. What does $S(10)$ mean in words?
(b) Describe the shape of the curve. Is $S(t)$ increasing or decreasing, and why?
(c) A constant hazard $h(t) = \lambda$ implies $H(t) = \lambda t$ and hence $S(t) = e^{-\lambda t}$. Verify this using the relationships in §1.4, and explain why a constant hazard gives this exponential shape.

Solution.

(a) $S(10) = e^{-1} \approx 0.37$ and $S(20) = e^{-2} \approx 0.14$; $S(10)$ is the probability of surviving beyond time 10.
(b) $S(t)$ is non-increasing and tends to 0: the probability of surviving beyond t can only fall as t grows.
(c) With $h(t) = \lambda$, $H(t) = \int_0^t \lambda \, \mathrm{d}u = \lambda t$, so $S(t) = \exp\{-H(t)\} = e^{-\lambda t}$ — a decreasing exponential curve.

3 Mini-lab: simulating survival and censoring

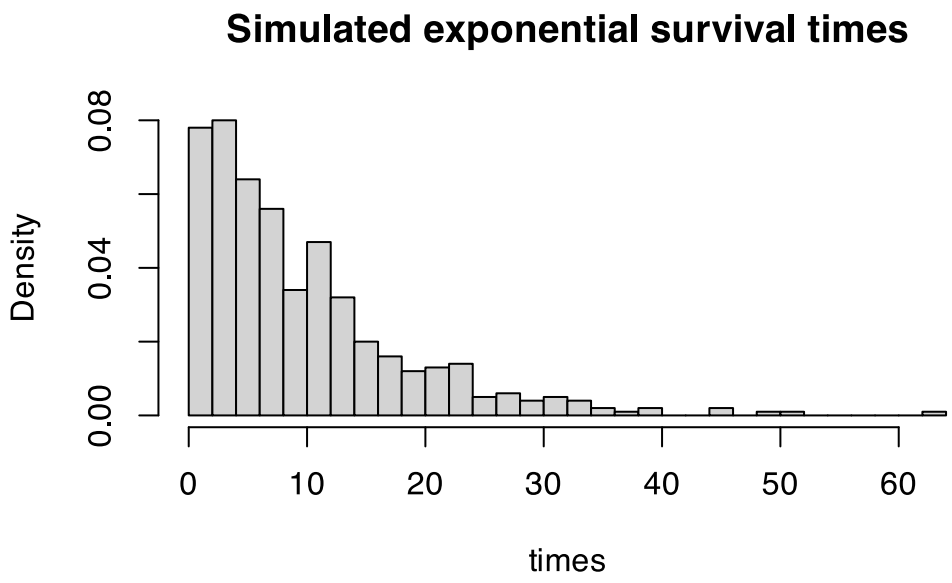
3.1 Simulating a censored sample

Type: Mini-lab

Run the code below. It simulates $n = 500$ exponential survival times with hazard $\lambda = 0.1$, and then introduces independent censoring times $C \sim \text{Uniform}(0, 15)$.

```r
set.seed(1)
n <- 500
lambda <- 0.1
times <- rexp(n, rate = lambda)

hist(times, probability = TRUE, breaks = 30,
     main = "Simulated exponential survival times")
```




```r
c <- runif(n, 0, 15)
censored <- c < times
cat("Censoring proportion:", round(mean(censored), 3), "\n")
```

```
Censoring proportion: 0.526
```

(a) The histogram is positively skewed, with a long right tail. Why is this typical of survival data (§1.2)?

(b) For an exponential survival time the hazard is constant, $h(t) = \lambda$. Use the relationships in §1.4 to find $S(t)$ and $H(t)$.

(c) What proportion of the observations are censored, and what does "censored" mean for one of these observations?

Solution.

(a) Survival times are generally not symmetric: there is a long tail of individuals who survive a long time, so the histogram is right-skewed and a normal model is therefore inappropriate.

(b) $H(t) = \int_0^t \lambda \, \mathrm{d}u = \lambda t$ and $S(t) = \exp\{-H(t)\} = e^{-\lambda t}$.

(c) Roughly half (about 0.52, as printed): for those observations we only know that the survival time exceeds the generated censoring time, not its exact value.